Algebra 2 — Unit 5: Radical Functions

Practice Test — Study Copy — Answer Key

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Index **C**
2. Radicand **A**
3. Principal root **F**
4. Conjugate **H**
5. Extraneous solution **E**
6. Composition **B**
7. Inverse function **G**
8. Anchor point **D**

(A) Everything written underneath the radical symbol; when the index is even, it may not be negative.

(B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$.

(C) The small number in the crook of the radical symbol that says which power is being undone.

(D) The point (h, k) that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd.

(E) A candidate that satisfies the powered-up equation but fails when it is tested in the original.

(F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand.

(G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$.

(H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. Written with a rational exponent, $\sqrt[3]{x^5}$ is **(A)** $x^{5/3}$ ← (B) $x^{3/5}$ (C) x^{15} (D) $x^{-3/5}$
2. The domain of $f(x) = \sqrt{7 - 2x}$ is (A) $x \geq \frac{7}{2}$ (B) $x \geq -\frac{7}{2}$ **(C)** $x \leq \frac{7}{2}$ ← (D) all real numbers
3. The graph of $y = \sqrt{3x - 12} + 5$ begins at the point (A) $(12, 5)$ (B) $(4, -5)$ (C) $(-4, 5)$ **(D)** $(4, 5)$ ←
4. In simplest form, $\sqrt{75} + \sqrt{12}$ is (A) $\sqrt{87}$ **(B)** $7\sqrt{3}$ ← (C) $7\sqrt{6}$ (D) $10\sqrt{3}$
5. Which equation has *no real solution*?
   (A) $\sqrt[3]{x+1} = -4$ (B) $\sqrt{x-1} = 7$ **(C)** $\sqrt{x+6} = -2$ ← (D) $\sqrt{x} = 0$
6. If $f(x) = \sqrt{x}$ and $g(x) = x + 9$, then $(f \circ g)(x) =$ (A) $\sqrt{x} + 9$ (B) $\sqrt{9x}$ (C) $x\sqrt{x} + 9\sqrt{x}$ **(D)** $\sqrt{x+9}$ ←

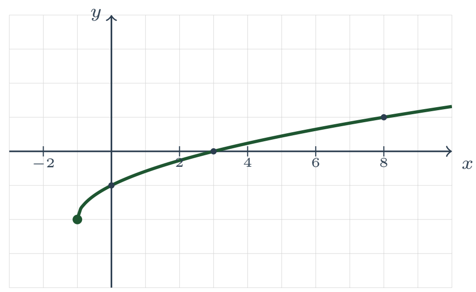
Teacher Note

Item 1: **denominator = index, numerator = power**. (B) is the signature Unit 5 error — the flipped fraction — and (C) multiplies index by power. Item 2: set the *radicand* ≥ 0: $7 - 2x \geq 0$ gives $-2x \geq -7$, and dividing by -2 **flips** the inequality. (A) is the un-flipped answer. Item 3: factor first — $\sqrt{3x-12} = \sqrt{3}\sqrt{x-4}$, so the graph starts at $x = 4$, not $x = 12$; (A) is the students who never factored. Item 4: $5\sqrt{3} + 2\sqrt{3}$; (A) adds radicands, (C) multiplies them, (D) multiplies the coefficients. Item 5: an isolated *even* root can never equal a negative number. Distractor (A) is the one to sort for — a student who learned "roots can't be negative" instead of "*even* roots can't" picks it, and that is a whole-unit failure, not a slip; $\sqrt[3]{x+1} = -4$ solves fine ($x = -65$). Item 6: the function next to the x runs first. (A) is the order error, (C) reads $\circ$ as multiplication.

Part C — Short Answer & Computation (40 pts)

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function f is graphed below; each gridline is one unit and the solid dot is the endpoint.



(a) Coordinates of the endpoint: $(-1, -2)$

(b) Domain (interval notation): $[-1, \infty)$; range: $[-2, \infty)$

(c) x-intercept: $(3, 0)$; y-intercept: $(0, -1)$

(d) Interval(s) where f is increasing: $[-1, \infty)$ — **the whole graph**

(e) Does f have an absolute maximum or an absolute minimum? Give it and say where it occurs.

An absolute minimum of -2, at the endpoint $x = -1$. There is no maximum: the arm climbs forever.

(f) End behavior: as $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, **nothing — the graph stops at $x = -1$**

(g) Interval(s) where $f(x) < 0$: $[-1, 3)$

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

(a) Write $\sqrt[4]{x^3}$ with a rational exponent. $x^{3/4}$

(b) Write $y^{5/2}$ as a radical. $\sqrt{y^5}$**, or** $\left(\sqrt{y}\right)^5$

(c) $27^{2/3} = 9$ (d) $16^{-1/2} = \frac{1}{4}$

(e) Write $x^{1/2} \cdot x^{1/3}$ as a single radical. **Add the exponents: $x^{1/2+1/3} = x^{3/6+2/6} = x^{5/6} = \sqrt[6]{x^5}$. As radicals the two factors could not be combined at all — mismatched indices — which is exactly why the rational-exponent notation exists.**

3. **Simplify. Add or subtract where you can.**

(a) $\sqrt{72}$ (b) $\sqrt[3]{54}$

(a) Largest perfect square: $\sqrt{36 \cdot 2} = 6\sqrt{2}$. (Stopping at $2\sqrt{18}$ is true but unfinished — always check the leftover.)

(b) The index sets the group size, so hunt for a perfect *cube*: $\sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$.

(c) $\sqrt{20x^3}$ (d) $5\sqrt{12} - \sqrt{27}$

(c) $\sqrt{4x^2 \cdot 5x} = 2x\sqrt{5x}$ — divide the exponent by the index: x^3 gives one x out and one x left in.

(d) Simplify first, decide second: $5 \cdot 2\sqrt{3} - 3\sqrt{3} = 10\sqrt{3} - 3\sqrt{3} = 7\sqrt{3}$. Before simplifying these look like unlike radicals.

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a) $\sqrt{6}\cdot\sqrt{15}$ (b) $\left(2+\sqrt{5}\right)\left(2-\sqrt{5}\right)$

(a) $\sqrt{90}=\sqrt{9\cdot 10}=3\sqrt{10}$. Neither factor simplifies on its own — the *product* does, so the last step is the graded one.

(b) Conjugates: $4-5=-1$, a rational number. Compare $\left(2+\sqrt{5}\right)^2=9+4\sqrt{5}$, where the radical survives.

(c) $\dfrac{6}{\sqrt{3}}$ (d) $\dfrac{4}{3-\sqrt{5}}$

(c) Multiply by $\frac{\sqrt{3}}{\sqrt{3}}=1$: $\dfrac{6\sqrt{3}}{3}=2\sqrt{3}$.

(d) Multiply by the *conjugate* $\frac{3+\sqrt{5}}{3+\sqrt{5}}$: $\dfrac{4\left(3+\sqrt{5}\right)}{9-5}=\dfrac{4\left(3+\sqrt{5}\right)}{4}=3+\sqrt{5}$. Using $3-\sqrt{5}$ again would only produce a new radical.

5. **Transformations.** Let $g(x)=-2\sqrt{x+3}+4$.

(a) List every transformation that takes $y=\sqrt{x}$ to g. **left 3, vertical stretch by 2, reflection over the x-axis, up 4**

(b) Anchor point: $(-3,4)$

(c) Domain: $[-3,\infty)$; range: $(-\infty,4]$

(d) Is the anchor an absolute maximum or an absolute minimum? **maximum ($a<0$ turns the arm over)**

(e) A different graph of the form $y=\sqrt{x-h}+k$ begins at $(5,-2)$. Write its equation. $y=\sqrt{x-5}-2$

(f) For $p(x)=\sqrt[3]{x-1}+2$, give the anchor point $(1,2)$ and the domain $(-\infty,\infty)$. In one sentence, say why that domain is not like g's.

The index 3 is odd, and cubing keeps the sign, so a cube root forbids no input — there is nothing to solve, and the anchor is an inflection point rather than an endpoint.

6. **Solve. Test every candidate in the original equation.**

(a) $\sqrt{2x+3}=x$ **Square: $2x+3=x^2$, so $x^2-2x-3=0$ and $(x-3)(x+1)=0$, giving candidates $x=3$ and $x=-1$. Check $x=3$: $\sqrt{9}=3$ ✓. Check $x=-1$: $\sqrt{1}=1$, but the right side is -1, so $x=-1$ is extraneous. Solution: $x=3$ only. (Note the tell — the non-radical side is negative at the rejected value.)**

(b) $\sqrt[3]{x-4}=-2$ **Cube both sides — legal in both directions, since every real number has exactly one real cube root: $x-4=-8$, so $x=-4$. Check: $\sqrt[3]{-8}=-2$ ✓. No extraneous risk with an odd index.**

(c) $\sqrt{3x-2}+4=1$ **Isolate first: $\sqrt{3x-2}=-3$. An isolated *even* root is never negative, so no real solution — and no algebra was needed. (Squaring anyway gives $x=\frac{11}{3}$, a phantom that fails the check.)**

(d) $x^{2/3}=25$ **Raise both sides to the reciprocal power $\frac{3}{2}$: $x=\pm 25^{3/2}=\pm 125$. The *numerator* 2 is even — it is a square in disguise — so write the $\pm$ yourself. Both check: $(\pm 125)^{2/3}=(\pm 5)^2=25$ ✓**

7. **Composition.** Let $f(x)=\sqrt{x}$ and $g(x)=x-6$.

(a) $(f\circ g)(x)=\sqrt{x-6}$, domain: $x\geq 6$

(b) $(g\circ f)(x)=\sqrt{x}-6$, domain: $x\geq 0$

(c) Are these the same function? Answer using *both* the rule and the domain.

No. The rules differ — one shifts the graph right 6, the other drops it down 6 — and so do the domains: $x\geq 6$ against $x\geq 0$. Order is part of the answer.

(d) $(f\circ g)(10)=\sqrt{4}=2$

(e) For $p(x)=2x+1$ and $q(x)=x^2-3$, find $(p\circ q)(-2)$. $q(-2)=1$, **then** $p(1)=3$

8. **Inverses.**

(a) $f(x) = 4x - 7$. Find $f^{-1}(x)$. $f^{-1}(x) = \dfrac{x+7}{4}$

(b) Verify by composition — *both* orders — that your answer really is the inverse. $f\big(f^{-1}(x)\big) = 4 \cdot \dfrac{x+7}{4} - 7 = x + 7 - 7 = x$, **and** $f^{-1}(f(x)) = \dfrac{(4x-7)+7}{4} = \dfrac{4x}{4} = x$. **Both orders return** x, **so they are inverses. One order alone is not a verdict.**

(c) $g(x) = \sqrt{x+2}$. Find $g^{-1}(x)$ and state its domain. **Swap and solve:** $x = \sqrt{y+2}$, **so** $x^2 = y + 2$ **and** $y = x^2 - 2$. **Domain:** $x \geq 0$, **because that is the *range* of** g **— read it off, never guess it.**

(d) The point $(7, 3)$ lies on the graph of g. What point must lie on the graph of g^{-1}? $(3, 7)$

Teacher Note

Item 1(f) is the Unit 5 habit-breaker: "the graph stops" is a *complete* answer to the left-end question, not a missing one — do not accept "$-\infty$" or a blank. Item 1(g): the endpoint is included, so $[-1, 3)$; a student writing $(-1, 3)$ has forgotten that $\sqrt{0}$ is real. Item 2(e): $x^{5/6}$ alone earns most of the credit, but the item asked for a radical. Item 3: an answer that is simplified but not *fully* simplified (e.g. $2\sqrt{18}$) is half an answer. Item 4(d): watch for the $a^2 - b^2$ sign flip in the denominator. Item 5(d) is the most-missed — a negative a makes the anchor a **maximum**, and the range flips with it; 5(f) must name the *index*, not just assert "all reals." Item 6(a) is the graded check: "$x = 3$ and $x = -1$" as a final answer earns no credit, because the check is the step being assessed — and 6(c) should be answered *before* any squaring. Item 7(c) must cite the domain, not only the rule; "they look different" is not the argument. Item 8(b) is A2.F.2i's justification clause — both orders, or it is not a verification.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \sqrt{x-1} + 2$. **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the x-intercept and the y-intercept, or explain why one does not exist. **(c)** List the transformations that take $y = \sqrt{x}$ to f. **(d)** Find $f^{-1}(x)$ and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that f and f^{-1} are inverses. **(f)** A classmate writes $y = (x-2)^2 + 1$ with *no* restriction and calls it the inverse. Explain why that is not the inverse of f. **(a) Anchor** $(1, 2)$; **domain** $[1, \infty)$; **range** $[2, \infty)$. **Both start at the anchor.**

   (b) Neither exists. An x**-intercept would need** $\sqrt{x-1} + 2 = 0$, **i.e.** $\sqrt{x-1} = -2$, **and an even root is never negative. A** y**-intercept would need** $f(0)$, **but** 0 **is not in the domain — the graph does not reach the** y**-axis.**

   (c) Right 1, **up** 2. ($a = 1$: **no stretch, no reflection.)**

   (d) Swap and solve: $x = \sqrt{y-1} + 2$, **so** $x - 2 = \sqrt{y-1}$, $(x-2)^2 = y - 1$, **and** $f^{-1}(x) = (x-2)^2 + 1$ **for** $x \geq 2$. **The restriction is the range of** f, **written down: whatever** f **can output is exactly what** f^{-1} **can accept.**

   (e) $f\big(f^{-1}(x)\big) = \sqrt{(x-2)^2 + 1 - 1} + 2 = \sqrt{(x-2)^2} + 2 = |x-2| + 2$, **and since** $x \geq 2$ **that is** $(x-2) + 2 = x$. **Going the other way,** $f^{-1}(f(x)) = \left(\sqrt{x-1} + 2 - 2\right)^2 + 1 = \left(\sqrt{x-1}\right)^2 + 1 = (x-1) + 1 = x$ **for** $x \geq 1$. **Both orders return** x, **so the two functions undo each other.**

   (f) Without the restriction it is the *whole* parabola — two arms — and it fails the horizontal line test, so it cannot be the inverse of anything: it sends both $x = 0$ **and** $x = 4$ **to** 5. **It also accepts inputs** f **never produces;** f **never outputs a number below** 2, **so** $x = 0$ **has no business being fed to** f^{-1}. **Only the right arm,** $x \geq 2$, **is the inverse.**

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops h feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

   (a) State the domain that makes sense here, and connect it to the rule about what an even index will accept. **(b)** Find $v(9)$ and $v(36)$, with units. What does quadrupling the drop do to the speed, and which feature of $\sqrt{\cdot}$ is responsible? **(c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer. **(d)** Write the inverse, h as a function of v, and say what it tells an engineer that $v(h)$

does not. **(e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem. **(a) $h \geq 0$, i.e. $[0, \infty)$. The index is even, so the radicand may not be negative — and the context says the same thing twice over, since a drop of negative height is not a thing. Here the algebra and the situation agree; that is not always true.**

(b) $v(9) = 8\sqrt{9} = 8 \cdot 3 = 24$ feet per second; $v(36) = 8\sqrt{36} = 8 \cdot 6 = 48$ feet per second. Quadrupling the drop only doubles the speed, because $\sqrt{4h} = 2\sqrt{h}$ — an exponent of $\frac{1}{2}$ flattens growth, which is the shape of the square root graph.

(c) $8\sqrt{h} = 56$, so $\sqrt{h} = 7$ and $h = 49$ feet. Check: $v(49) = 8\sqrt{49} = 8 \cdot 7 = 56$ ✓

(d) Solve for h: $\sqrt{h} = \frac{v}{8}$, so $h = \dfrac{v^2}{64}$ for $v \geq 0$. The original answers "how fast will this drop make the car go?"; the inverse answers the question an engineer actually asks — "how tall must I build the drop to hit a target speed?"

(e) An extraneous root only appears when squaring hides a *sign* disagreement — the step $a^2 = b^2$ forgets whether $a = b$ or $a = -b$. Here both sides are non-negative: $\sqrt{h} \geq 0$ and a speed of 56 is positive, so there is no hidden sign to lose and the single candidate must be genuine. Checking it is still good practice.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) anchor with *both* interval-notation answers; 1 pt (b) *both* non-existences with reasons — "no x-intercept" with no reason earns half; 1 pt (c) both transformations; 1 pt (d) the rule *and* $x \geq 2$ *and* the sentence that it is the range of f (no credit for the restriction if it is guessed as $x \geq 0$, the reflex answer); 1 pt (e) both compositions — one order is not a verification, and this is the standard's own "justify" clause; 1 pt (f) the horizontal line test or the two-arms argument. A student who writes $(x-2)^2 + 1$ with no restriction in (d) and then cannot explain (f) has missed the unit's closing idea — score both accordingly. D2: 1 pt (a) $h \geq 0$ tied to the even index; 1 pt (b) both values with units *and* "doubles" with $\sqrt{4h} = 2\sqrt{h}$ (half credit for "it doubles" with no reason); 1 pt (c) $h = 49$ ft; 1 pt the check; 1 pt (d) $h = \frac{v^2}{64}$; 1 pt (e) the sign argument. Do not accept "because I checked it" for (e) — the question asks why the danger was never there.